

Q5) Smart Campus Parking and Access Management System.

(1) PAC Chart

* Given Data

Zone A $\rightarrow$ max = 20

Zone B $\rightarrow$ max = 40

Zone C $\rightarrow$ max = 15

Vehicle type (C, B, V)

User category (F, S, G)

* Processing required

$\rightarrow$ Validate inputs

• Check capacity

•

* Required Results

• For every successfully parked vehicle system must display assigned zone & remain. capacity

• For rejected $\rightarrow$ reason for rejected
e.g. invalid permit

* Solution alternatives

(2) IPO Chart

Input

- No of vehicles
- Vehicle type
- user category
- Parking permit

Process

- ~~Validate~~
- Validate inputs and reject vehicles without permit, except if it's an emergency vehicle.

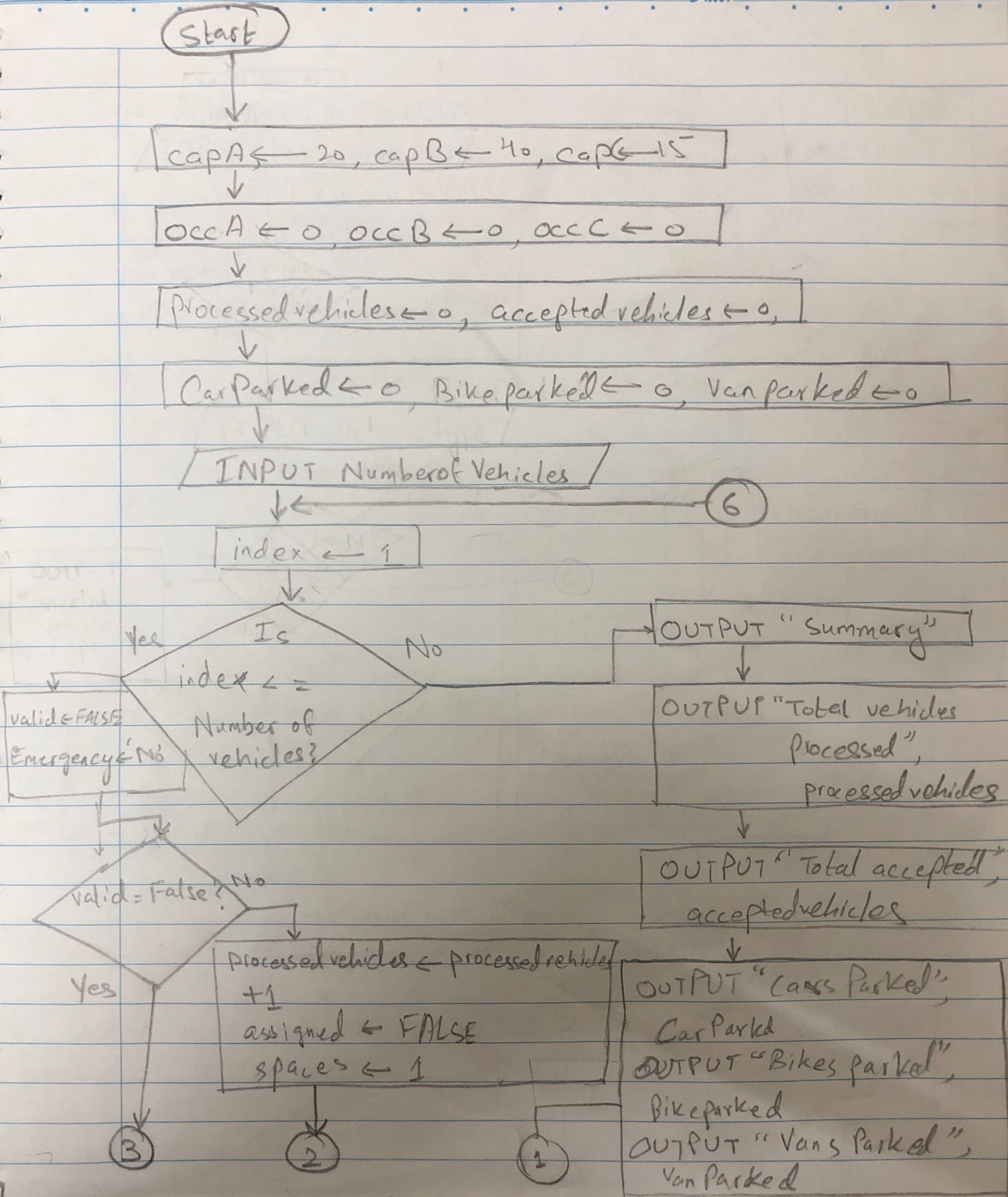
Output

Parking summary:

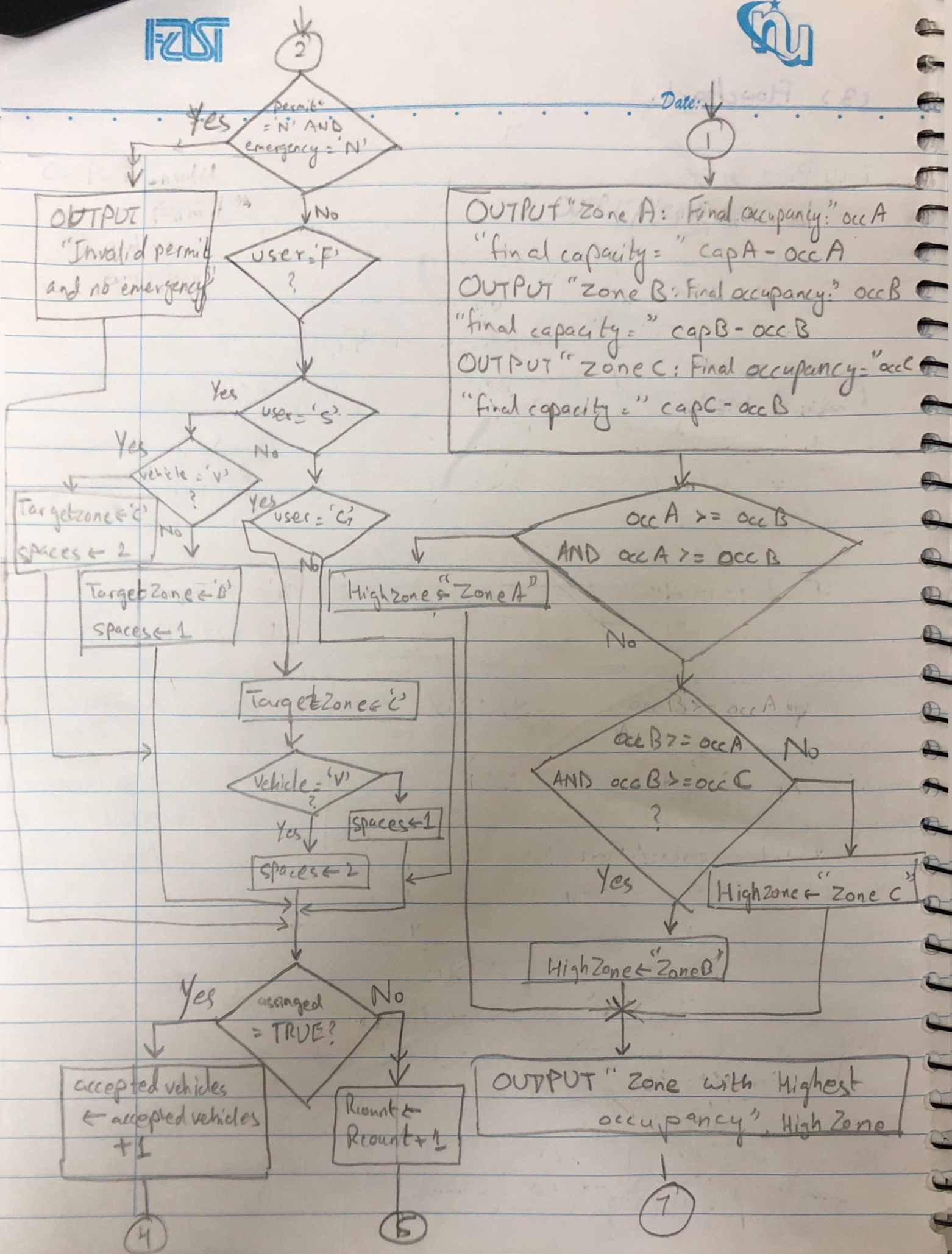
- Total number of vehicles processed
- Total accepted & rejected vehicle
- no. of cars, ~~vehicles~~ bikes
- vans successfully parked, final occupancy.
- remaining capacity
- Display zone with highest occupancy.
- is entire parking facility full

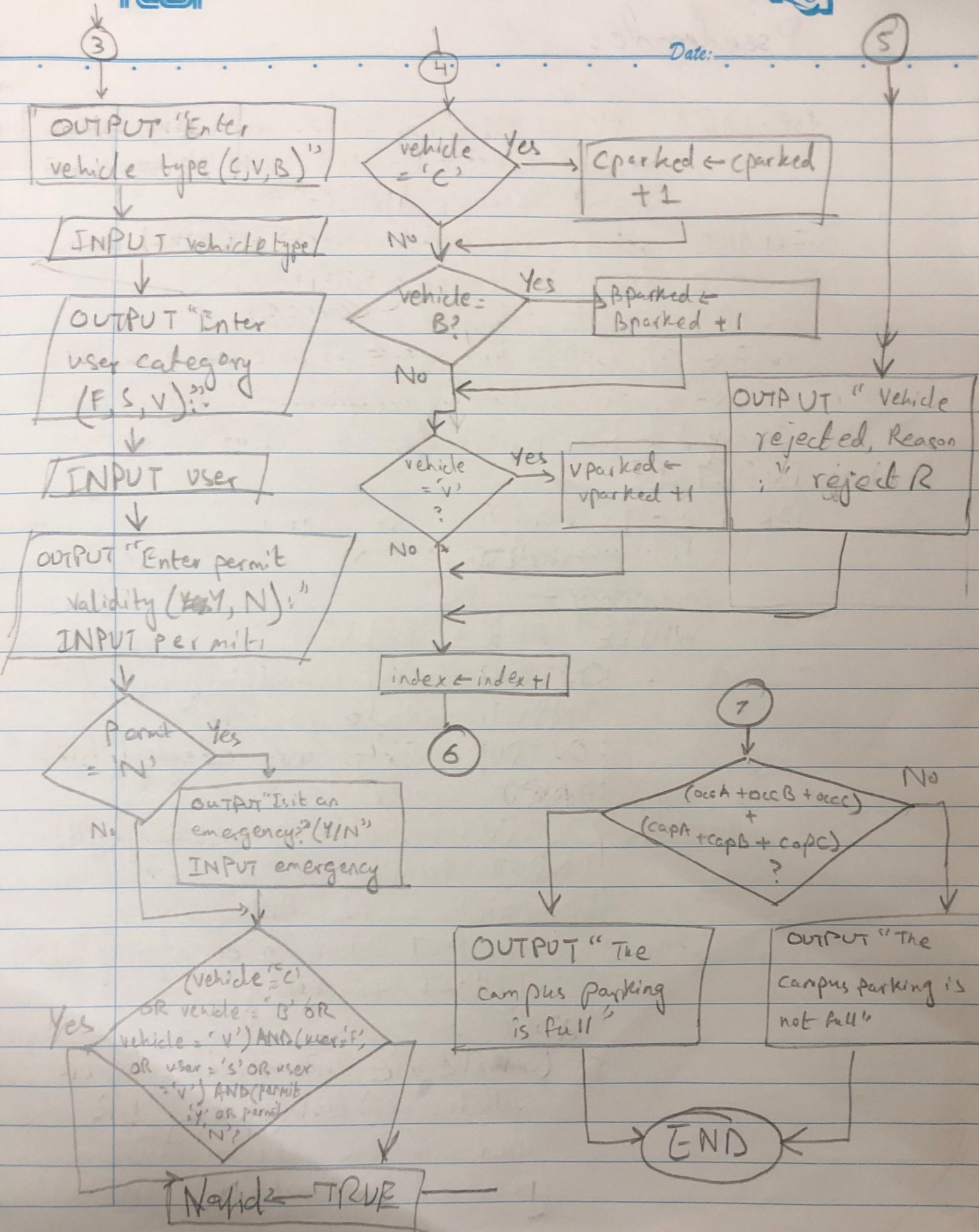
(3) Flowchart.

Date: _____



Date: 14/07/2019





Pseudocode:

Date:

```

DECLARE N, i, capA, capB, capC, occA, occB, occC,
        processed, accepted, Rcount, Cparked, Bparked,
        Vparked, spaces: INTEGER
DECLARE vehicle, user, permit, targetz: CHAR
DECLARE rejectR, Hzone: STRING
DECLARE valid, assigned: BOOLEAN
capA ← 20, capB ← 40, capC ← 15, occA ← 0, occB ← 0,
occC ← 0, processed ← 0, accepted ← 0, rejected ← 0
Cparked ← 0, Bparked ← 0, Vparked ← 0
INPUT N
FOR i ← 1 TO N
    valid ← FALSE
    emergency ← 'N'
    WHILE valid = FALSE
        OUTPUT "Enter vehicle type (C/B/V): "
        INPUT vehicle
        OUTPUT "Enter user category (F/S/V): "
        INPUT user
        OUTPUT "Enter Permit validity (Y/N): "
        INPUT permit
        IF permit = 'N' THEN
            OUTPUT "Is it an emergency? (Y/N)"
            INPUT emergency
        ENDEF
        IF (vehicle = 'C' OR vehicle = 'B' OR
            vehicle = 'V') AND (user = 'F' OR
            user = 'S' OR user = 'V') AND
            (permit = 'Y' OR permit = 'N') THEN

```


Date: _____

valid = TRUE

ENDIF

ENDWHILE

processed $\leftarrow$ processed + 1

assigned $\leftarrow$ FALSE

spaces $\leftarrow$ 1

IF permit = 'N' AND emergency = 'N' THEN
OUTPUT "Invalid permit and no emergency"

ELSE

IF user = 'F' THEN

Target Z $\leftarrow$ 'A'

spaces $\leftarrow$ 1

ELSE

IF user = 'S' THEN

IF vehicle = 'V' THEN

Target Z $\leftarrow$ 'C'

spaces $\leftarrow$ 2

ELSE

Target Z $\leftarrow$ 'B'

spaces $\leftarrow$ 1

ENDIF

ELSE

IF user = 'C' THEN

IF vehicle = 'V' THEN

spaces $\leftarrow$ 2

ELSE

space $\leftarrow$ 1

ENDIF

ENDIF

IF Target Z = 'A' THEN

IF $(capA - occA) \geq spaces$ THEN

$occA \leftarrow occA + spaces$

assigned $\leftarrow$ TRUE

reject R $\leftarrow$ "Assigned to Zone A.

Remaining capacity = ", $(capA - occA)$

ELSE

OUTPUT "No available space
in Zone A"

ENDIF

ELSE

IF Target Z = 'B' THEN

IF $(capB - occB) \geq spaces$ THEN

$occB \leftarrow occB + spaces$

assigned $\leftarrow$ TRUE

OUTPUT "Assigned to Zone B.

Remaining capacity: "

$(capB - occB)$

ELSE

reject R $\leftarrow$ "No available space
in Zone B."

ENDIF

ELSE

IF Target Z = 'C' THEN

IF $(capC - occC) \geq spaces$ THEN

$occC \leftarrow occC + spaces$

assigned $\leftarrow$ TRUE

OUTPUT "Assigned to Zone C.

Remaining capacity: ", $(capC - occC)$

ELSE

Every MUN Topic is a problem



~~Understand the problem~~

~~and~~

~~Understand it well.~~

reject $R \leftarrow$ "No available space or suitable zone".

ENDIF

ENDIF

ENDIF

IF assigned = 'TRUE' THEN

accepted $\leftarrow$ accepted + 1

IF vehicle = 'C' THEN

Cparked $\leftarrow$ Cparked + 1

ENDIF

IF vehicle = 'B' THEN

Bparked $\leftarrow$ Bparked + 1

ENDIF

IF vehicle = 'V' THEN

Vparked $\leftarrow$ Vparked + 1

ENDIF

ELSE

Rcount $\leftarrow$ Rcount + 1

OUTPUT "Vehicle rejected. Reason: " reject-R

ENDIF

NEXT

OUTPUT "Summary:"

Date:

OUTPUT "Total vehicles processed: " processed

OUTPUT "Total Accepted: " accepted

OUTPUT "Total Rejected: " Rcount

OUTPUT "Cars Parked: " Cparked

" " Bikes Parked: " Bparked

" " Vans parked: " Vparked

OUTPUT "Zone A --> Final Occupancy: " occA

" " Remaining Capacity: (capA - occA)

OUTPUT "Zone B --> Final Occupancy: " occB

" " Remaining Capacity: (capB - occB)

OUTPUT "Zone C --> Final Occupancy: " occC

" " Remaining Capacity: (capC - occC)

IF $occA \geq occB$ AND $occA \geq occC$ THEN

Hzone $\leftarrow$ "Zone A"

ELSE

IF $occB \geq occA$ AND $occB \geq occC$ THEN

Hzone $\leftarrow$ "Zone B"

ELSE

Hzone $\leftarrow$ "Zone C"

ENDIF

ENDIF

OUTPUT "Zone with highest occupancy: " Hzone

IF $(occA + occB + occC) = (capA + capB + capC)$ THEN

OUTPUT "The entire campus parking is full"

ELSE

OUTPUT "The campus parking is not full"

ENDIF